

Maya Wallach

Curriculum Vitae

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SUMMARY

Physics/CMSE PhD candidate applying deep learning to scientific and real-time systems, with a parallel focus on low-level embedded and kernel-level programming.

EDUCATION

Michigan State University	East Lansing, MI
PhD Physics and Computational Mathematics, Science and Engineering (CMSE)	2024 – Present
BS Physics (Honors, Full Ride)	2021 – 2024

SKILLS

Languages: Python, C/C++, Verilog, VHDL, CUDA, Rust, SQL, R, x86 Assembly
ML/Scientific: PyTorch, TensorFlow, scikit-learn, NumPy, pandas, OpenMPI
Systems & Tools: Git, Linux, SLURM, CMake, Make, GDB, OpenOCD, TCP/IP networking
Hardware: STM32 (Cortex-M4), ARM architecture, KiCad, SWD/JTAG

EXPERIENCE

Graduate Research Assistant, Michigan State University	2024 – Present
- Used machine learning methods for extracting weak signals from noisy datasets using deep neural networks and probabilistic modeling in PyTorch and Tensorflow - Developed neural network-based domain adaptation methods to reduce simulation-to-data distribution shifts in Monte Carlo-generated datasets - Optimized a transformer-based object detection model (RT-DETR) via quantization and pruning, improving inference throughput by 45% on a 4k+ image dataset for real-time, compute-constrained deployment - Implemented DINOv3 SSL pretraining for the RT-DETR backbone to reduce reliance on labeled data - Profiled and benchmarked model performance across hardware configurations, generating telemetry and latency reports to guide optimization decisions - Collaborated across multiple research groups and mentored undergraduate students	
Undergrad Research Assistant, Fermi National Accelerator Laboratory	2023 – 2024
- Worked with an advisor to apply deep learning with PyTorch for particle identification, developing classifiers achieving 92% accuracy - Implemented a Faster R-CNN model using Detectron2 for object detection-based particle identification - Applied GAN-based (DCGAN) generative modeling to mitigate limited labeled training data	
IRIS-HEP Undergrad Fellow, Davidson College	2022
- Developed a self-supervised, semi-supervised event classification pipeline for the Active-Target Time Projection Chamber (AT-TPC) at FRIB, reducing dependence on manually labeled training data - Applied a PointNet-based architecture to 3D point-cloud particle trajectory data to classify nuclear reaction scattering events, exploiting order and transformation-invariance properties of point sets	
Undergrad Research Assistant, Facility for Rare Isotope Beams	2021 – 2024
- Built regression models for sparse-data prediction of unstable nuclear radii, developing techniques for inference under data constraints and high measurement uncertainty - Reimplemented digital and analog circuits used at FRIB on FPGAs, primarily clocking circuits	
Undergrad Research Intern, Los Alamos National Laboratory	2021 – 2022
- Implemented numerical fluid dynamics simulations for the k-l turbulence model in C/C++ - Optimized performance for scientific workloads on HPC systems	

CONFERENCE PRESENTATIONS

<i>Author</i> Estimating charge radii for rare isotopes from stable nuclei — APS	2023
<i>Author</i> The 2023 Physicists Inspiring the Next Generation: Exploring the Nuclear Matter — APS	2023
<i>Poster</i> Unsupervised Learning to Build Pretrained Models for the AT-TPC — DNP	2022

SELECTED PERSONAL PROJECTS

Deep Q-Network Agent for Warhammer 40k — Python, C++	GitHub Repo
Designed and implemented an end-to-end reinforcement learning framework for a custom stochastic strategy game, including environment simulation, reward shaping, experience replay, and Deep Q-Network training	

- Developed a custom Gymnasium-compatible environment and game engine to support autonomous agent training and evaluation
- Built a C++/gtkmm desktop application for interactive visualization, human-vs-agent gameplay, and model evaluation

Deep Learning Time Series Forecasting Framework — Python

GitHub Repo

- Researched cross-sectional market anomalies in equity price and volume data, constructing and testing systematic trading signals across 9 assets using SVM, RF, XGBoost, and LSTM models
- Built a transaction-cost-adjusted backtesting framework; evaluated strategies using Sharpe ratio, max draw-down, and t-statistics on out-of-sample returns via walk-forward validation across 5 test periods
- MPI-parallelized training reduced runtime by 95% versus sequential baseline, enabling rapid iteration over signal hypotheses and model variants

KlaudOS (Operating System from Scratch) — C, x86 assembly

GitHub Repo

- Designed and implemented a 32-bit x86 operating system in C and x86 assembly, including bootloader, kernel initialization, interrupt handling, and hardware drivers
- Configured processor structures (GDT, IDT, PIC) and wrote interrupt service routines to support keyboard input and system services
- Implemented a custom filesystem layer with disk I/O support and automated deployment/testing using QEMU and a custom build system

KlaudSynth (Embedded Audio Synthesizer) — C

GitHub Repo

- Built a real-time digital synthesizer in C targeting ALSA on Linux; managed audio buffer callbacks under strict latency constraints with explicit thread priority control and Linux kernel scheduling APIs
- Designed an STM32F407-based hardware version in KiCad: mixed-signal PCB with separate analog/digital power planes, MIDI optoisolation, 32-channel multiplexed ADC input, and OPA2134 reconstruction filter; firmware drives a 12-bit DAC via DMA circular buffer at 48kHz; validated hardware bring-up using SWD

KlaudScheduler (MPI Cluster Resource Manager) — C

GitHub Repo

- Designed a distributed resource manager for a personal Beowulf cluster, implementing a job queue and MPI-based dispatch system across head and compute nodes
- Built inter-node communication via TCP for node discovery and resource reporting, generating a dynamic inventory of available CPUs across the cluster
- Implemented job submission CLI and status tracking to manage concurrent workloads with configurable core and GPU allocation

AWARDS

NatSci SMART AI Fellowship (2026)
Thomas Osgood Award (2024)

University Distinguished Fellowship (2024–Present)
FRIB Merit Scholarship (Full Ride, 2021–2024)

LEADERSHIP & OUTREACH

National Society of Black Physicists - Secretary & Founding Member
Women and Minorities in the Physical Sciences - Treasurer
PING Mentor (Summer 2023)